

Week 4 Interim Submission

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GitHub link: [zaria1108/carisurg-portfolio: CariSurg Internship Week 0](https://github.com/zaria1108/carisurg-portfolio)

Draft Risk Register: Explainable AI-Assisted Emergency Department Triage Support System

Purpose

This risk register identifies initial risks associated with deploying an AI-assisted Emergency Department triage support system across public healthcare settings. Risks are grouped across technical, operational, ethical, and equity domains.

Risk Name	Category	Likelihood	Impact	Mitigation	Signal of Success
Recommendation Instability from Progressive Data Entry	AI-Technical	High	High	Delay final recommendations until minimum observations are available and continuously update outputs	Reduction in recommendation changes after complete triage
Context Loss Across Workflow Handoffs	Operational	High	High	Preserve structured summaries across registration, triage, and clinician review	Reduction in repeated information collection
Alert Fatigue During Queue Monitoring	Operational	Medium	High	Restrict alerts to escalation thresholds and suppress repetitive notifications	Sustained clinician response to reassessment alerts
Automation Bias During Triage Support	Ethical	Medium	High	Require clinician confirmation and display supporting factors for recommendations	Low override acceptance without independent review
Unequal Performance Across Hospital Sites	Equity	Medium	High	Validate models separately across hospitals before rollout	Comparable performance across deployment sites

Risk Memo - Highest Priority Risks

➤ Risk 1: Recommendation Instability from Progressive Data Entry

Within the Emergency Department workflow, information becomes available progressively rather than simultaneously. Registration occurs before observations and full clinical assessment are completed. If recommendations are generated too early, urgency support may change substantially after additional observations become available. This creates inconsistency and may reduce clinician trust.

Mitigation: Require minimum observation thresholds before generating recommendations and refresh outputs dynamically.

➤ Risk 2: Context Loss Across Workflow Handoffs

Patient information moves across administrative staff, triage nurses, waiting stages, and clinicians. Important details collected early may not remain equally visible later in the workflow. AI systems that fail to preserve context risk producing incomplete recommendations.

Mitigation: Carry structured summaries forward rather than requiring repeated data collection.

➤ Risk 3: Automation Bias During Triage Support

Clinicians working under operational pressure may gradually rely on AI recommendations without applying independent judgement. This risk increases when outputs appear precise or are presented without uncertainty indicators.

Mitigation: Display confidence information, preserve clinician authority and require confirmation before recommendation acceptance.

Documented AI Harm Case: Root Cause Notes

Case: Epic Sepsis Prediction Model

What Happened?

A machine-learning sepsis prediction system was deployed across hospitals to identify patients at elevated risk of developing sepsis. Independent evaluation later found that the deployed system identified substantially fewer cases than expected and generated limited practical benefit during clinical use.

Root Cause Analysis?

The system performed differently during deployment than anticipated from retrospective testing. Important contributing factors included differences between training conditions and real-world workflow conditions, delayed data availability, and assumptions about how clinicians would act on alerts.

What Failed?

- Performance did not generalize as expected
- Operational assumptions were not validated
- Monitoring after deployment was insufficient

What Would Have Reduced Harm?

- Prospective validation
- Site-level evaluation
- Continuous monitoring after rollout
- Human review before escalation

Relevance to This Proposal

This case reinforces that successful healthcare AI depends not only on predictive performance but also on workflow fit, data readiness, monitoring, and responsible clinical oversight.